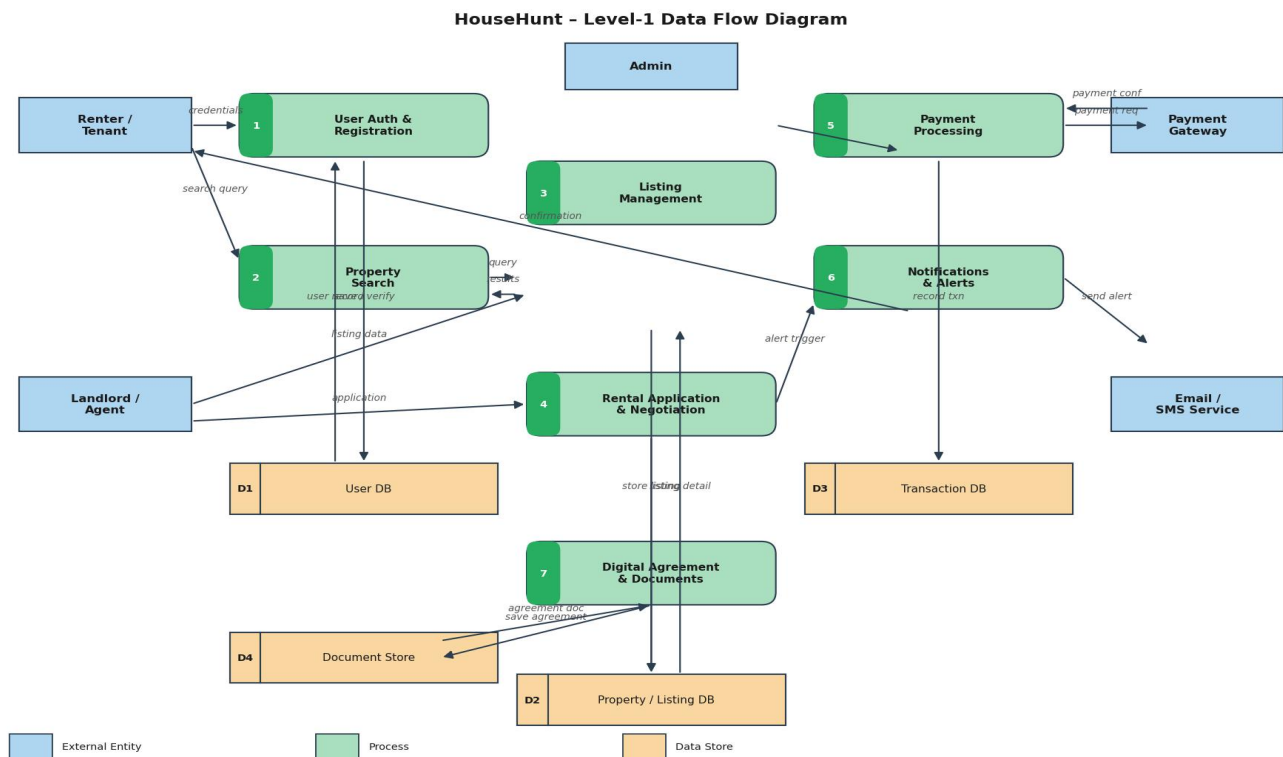


Project Design Phase-II Data Flow Diagram & User Stories

Date	18 June 2026
Project Name	HouseRental
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored. The Level-1 DFD below illustrates the key data flows within the HouseHunt house-rental platform, covering user authentication, property search and listing management, rental applications, payments, notifications, and digital agreements.



DFD Flow Description:

1. Renters and Landlords / Agents interact with the system via the mobile / web app.
2. User Auth & Registration (P1) validates credentials and stores / retrieves user records from the User DB (D1).
3. Property Search (P2) accepts search queries from Renters and retrieves matching listings from the Property / Listing DB (D2).
4. Listing Management (P3) allows Landlords / Agents to create, update, or remove listings, which are persisted in the Property / Listing DB (D2).
5. Rental Application & Negotiation (P4) receives applications from Renters, updates application status, triggers Notifications & Alerts (P6), and feeds into Digital Agreement & Documents (P7).

6. Payment Processing (P5) communicates with the external Payment Gateway, records transactions in the Transaction DB (D3), and triggers payment-confirmation alerts.
7. Notifications & Alerts (P6) sends emails and SMS messages via the external Email / SMS Service to keep all parties informed.
8. Digital Agreement & Documents (P7) generates and stores signed rental agreements and KYC documents in the Document Store (D4).
9. Admin users oversee the system, approve listings, and access platform analytics.

User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	USN No.	User Story / Task	Acceptance Criteria	Priority	Release
Renter (Mobile / Web)	Registration & Authentication	USN-1	As a renter, I can register by entering my name, email, and password.	Account is created; I am redirected to my dashboard.	High	Sprint-1
		USN-2	As a renter, I receive an OTP on my registered mobile number to verify my account.	OTP validated; account marked as verified.	High	Sprint-1
		USN-4	As a renter, I can reset my password via a link sent to my registered email.	Password reset link received; new password accepted.	Medium	Sprint-2
	Property Search	USN-5	As a renter, I can search for properties by entering a city or locality.	Relevant listings appear within 2 seconds.	High	Sprint-1
		USN-6	As a renter, I can filter search results by rent range, number of bedrooms, and amenities.	Only listings matching all selected filters are displayed.	High	Sprint-1
		USN-7	As a renter, I can view properties on an interactive map.	Map renders with pins for each listing; clicking a pin shows a summary card.	Medium	Sprint-2
	Property Details & Virtual Tour	USN-8	As a renter, I can view full property details including photos, floor plan, and nearby amenities.	All details load on the listing page; images open in full-screen gallery.	High	Sprint-1
		USN-9	As a renter, I can watch a virtual / 360° tour of the property without visiting in person.	Video tour plays smoothly within the app on 4G connection.	Medium	Sprint-2
	Enquiry & Communication	USN-10	As a renter, I can send an enquiry message to the landlord / agent directly from the listing	Message delivered; landlord receives in-app and email	High	Sprint-1

			page.	notification.		
	Rental Application	USN-12	As a renter, I can submit a formal rental application with my personal and financial details.	Application submitted; status shows 'Under Review'.	High	Sprint-2
		USN-13	As a renter, I can track the status of my rental application in real time.	Status updates (Submitted → Under Review → Accepted / Rejected) visible on my dashboard.	High	Sprint-2
	Agreements	USN-15	As a renter, I can digitally sign the rental agreement within the app.	E-signed agreement saved to my account and landlord's account.	High	Sprint-3
	Ratings & Reviews	USN-16	As a renter, I can rate and review a property after my check-in.	Review visible on listing page; average rating updated.	Low	Sprint-4
Landlord / Agent	Listing Management	USN-17	As a landlord, I can create a new property listing with photos, description, rent, and amenities.	Listing submitted for admin review and published after approval.	High	Sprint-1
		USN-18	As a landlord, I can edit or pause my active listing at any time.	Changes saved and reflected on the listing page within 60 seconds.	High	Sprint-1
		USN-19	As a landlord, I can mark a property as 'Rented' to remove it from active search results.	Listing no longer appears in search; shown as 'Rented' in my dashboard.	Medium	Sprint-2
Agent / Broker	Agent Dashboard	USN-22	As an agent, I can manage all my property listings from a single dashboard.	All listings visible with enquiry count, application status, and conversion metrics.	High	Sprint-2
Administrator	Content Moderation	USN-24	As an admin, I can approve or reject property listings and KYC documents submitted by landlords and agents.	Listing status updated; landlord notified of approval or rejection with reason.	High	Sprint-1
		USN-25	As an admin, I can suspend or ban user accounts that violate platform policies.	Suspended user cannot login; receives an email explaining the action.	High	Sprint-1
	Platform Management	USN-27	As an admin, I can manage subscription plans and apply promotional discount codes.	Plan changes and coupon codes reflect immediately for users.	Medium	Sprint-4

